

X(5th Sm.)-Computer Sc.-G/SEC-A-1/CBCS

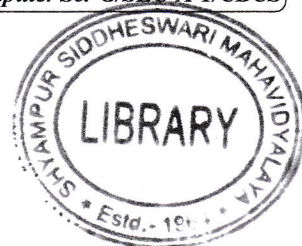
2022

COMPUTER SCIENCE — GENERAL

Paper : SEC-A-1

(Communication, Computer Network and Internet)

Full Marks : 80



The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question nos. 1 and 2** and **any four** questions from the rest.

1. Answer **any ten** questions :

2×10

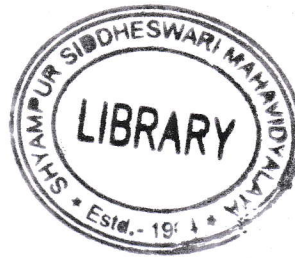
- (a) What is Baud rate?
- (b) What are the functions of a router?
- (c) Why is default gateway used?
- (d) How are TCP and UDP different?
- (e) What is channel capacity?
- (f) What is Geostationary satellite?
- (g) Mention two advantages of Star topology.
- (h) What are the drawbacks of client-server model?
- (i) State the range of Class B IP address.
- (j) What is MAC address?
- (k) What are the different layers of TCP/IP model?
- (l) What is attenuation?
- (m) What is URL?
- (n) What is NAT?
- (o) Mention few features of Delta modulation.

2. Write short notes on (**any four**) :

5×4

- (a) VSAT
- (b) QAM
- (c) ISDN

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(2)

- (d) OSI model
 - (e) FTP
 - (f) Amplitude Modulation.
3. (a) Briefly describe how FDM works.
- (b) Assume that a voice channel occupies a bandwidth of 4 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth.
- (c) What is frequency shift keying? 5+3+2
4. (a) State the differences between connectionless and connection oriented services.
- (b) Discuss different characteristics of WAN.
- (c) Explain Nyquist theorem. 4+3+3
5. (a) Encode the string 01001110 in NRZ-L and Differential Manchester encoding scheme. Clearly mention the conventions.
- (b) Compare between ASK and FSK.
- (c) What is bit length? 4+4+2
6. (a) What is SNR? Define its unit.
- (b) An analog signal carries 4 bits/signal element. If 1000 signal elements are sent per second, find the bit rate.
- (c) Mention advantages of optical fibre for data communication.
- (d) State the relationship between SNR and channel capacity of a noisy channel. (2+1)+2+3+2
7. (a) How are Public IP different from Private IP?
- (b) What are the basic functions of DNS server?
- (c) Briefly discuss the structure of a data packet. 3+2+5
8. (a) What are the differences between IPV4 and IPV6 addressing?
- (b) How does the web browser reach an IP address? Use schematic diagram to explain the process. 5+5
9. (a) What is search engine?
- (b) What is the purpose of multiplexing?
- (c) Briefly explain different services provided by the Application layer in internet model. 2+4+4
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